

Scalability & Future Plan

Date	08 October 2026
Team ID	UNM1611GUFRAN/SWTID-2026-9802
Project Name	AI FAQ Assistant API
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Runs on a single API instance with no load balancing	Slow responses or downtime during heavy traffic	High
2	FAQ answers depend on a fixed knowledge base	Answers may be outdated or missing for new topics	Medium
3	Supports English language only	Limits reach to non-English users	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single server instance	Add load balancing and auto-scaling on a cloud platform
2	Data Storage	Single local database	Move to a managed cloud database with regular backups
3	Performance	Each query processed directly	Add caching for frequent questions and asynchronous processing
4	Security	Basic API key access	Add token-based authentication, rate limiting and HTTPS encryption

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy API on cloud with load balancing	Nov 2026	Handles more users with stable uptime

Phase 2	Admin dashboard to add and update FAQs	Jan 2027	Easier knowledge base updates without code changes
Phase 3	Multilingual support (Tamil, Hindi and more)	Mar 2027	Reaches a wider audience
Phase 4	Usage analytics and web / WhatsApp chatbot integration	Jun 2027	Better insights and wider access for users